Arc Length and Improper Integrals

Section 1: Arc Length of a Plane Curve

1 State the formula for the arc length L of $y = f(x)$ from $x = a$ to $x = b$.

a. $L = \int_a^b \sqrt{1 + [f'(x)]^2}\, dx$

b. $L = \int_a^b \sqrt{1 + f'(x)}\, dx$

c. $L = \int_a^b [1 + f'(x)]\, dx$

d. $L = \int_a^b \sqrt{[f(x)]^2 + [f'(x)]^2}\, dx$

2 Find the arc length of $y = \frac{2}{3}x^{3/2}$ from $x = 0$ to $x = 3$.

a. $\frac{14}{3}$

b. 7

c. $\frac{16}{3}$

d. 8

3 Find the arc length of the line segment $y = \frac{4}{3}x + 2$ from $x = 0$ to $x = 3$.

a. 5

b. 4

c. 6

d. $\sqrt{7}$

Section 2: Improper Integrals of Type I (Infinite Bounds)

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4 Evaluate $\int_1^\infty \frac{1}{x^2}\,dx$.

a. 1

b. ∞ (diverges)

c. 1/2

d. 2

5 Evaluate $\int_0^\infty e^{-2x}\,dx$.

a. 1/2

b. 1

c. ∞

d. 2

6 For what values of p does the improper integral $\int_1^\infty \frac{1}{x^p}\,dx$ converge?

a. $p > 1$

b. $p \geq 1$

c. $p < 1$

d. $p > 0$

Section 3: Improper Integrals of Type II (Discontinuous Integrands)

7 Evaluate $\int_0^1 \frac{1}{\sqrt{x}}\,dx$.

a. 2

b. 1

c. ∞ (diverges)

d. 1/2

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8 Evaluate $\int_0^2 \frac{1}{x-1}\,dx$.

a. Diverges

b. 0

c. $\ln 2$

d. ∞

9 For what values of p does $\int_0^1 \frac{1}{x^p}\,dx$ converge?

a. $p < 1$

b. $p > 1$

c. $p \leq 1$

d. $p > 0$

Section 4: Comparison Test for Improper Integrals

10 Determine the convergence of $\int_1^\infty \frac{\sin^2(x)}{x^2}\,dx$.

a. Converges by comparison with $\frac{1}{x^2}$

b. Diverges

c. Oscillates without limit

d. Converges to 0

11 Determine the convergence of $\int_1^\infty \frac{1}{\sqrt{x^3+1}}\,dx$.

a. Converges (behaves like $x^{-3/2}$ as $x \to \infty$)

b. Diverges (behaves like $x^{-1/2}$)

c. Diverges to $-\infty$

d. Cannot be tested

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12 Evaluate $\int_{-\infty}^{\infty} \frac{1}{1+x^2}\,dx$.

a. π

b. 2π

c. $\pi/2$

d. Diverges

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